Intro to AI -- Main Project (2026)

The main project of the **Introduction to AI** course consists of a case study on how AI can be used to process a particular dataset.

In many scenarios, companies have access to datasets (for instance, sales, logistics records, IT bug reports and support tickets, etc.), but without much knowledge about how to extract relevant information from this data. Hence, automated extraction of information and patterns from data may appear as a very practical use case of AI. We propose to tackle such a scenario as the final evaluation of the class.

The project is the backbone of the course. You choose it in the first session, validate it in the second, and then apply every technique seen in the labs directly to your own project data. **Your project will be your application during the whole course**. Everything you do in the labs from Session 3 onward feeds your final project deliverable.

How the project fits into the course

When	What happens for the project
Session 1 (dedicated 2h30 slot)	**Define your own project**, based on what your binôme wants to work on and your prior knowledge. A good way to start is to look for a dataset and build a task around it. A list of ready-to-use projects is available as a backup. You also form your binôme and register it on Discord.
Before Session 2	Post your **project title** (one-liner) in your Discord channel for early validation by a teacher.
Session 2 (whole session)	**Validation presentation**: you present task, data, metrics and existing methods. Teachers validate. **Graded.**
Session 3 → onward	Each lab (Supervised, Unsupervised, Deep Learning, Foundation Models, Adaptation) will ask you to **apply techniques to your project data**, in addition to toy examples.
End of the course	Dedicated project work, then the **final presentation**.

A note on AI-usage

- You may use **any AI assistant** to help you (e.g., ChatGPT, Claude, ...), at any moment in the project. Note though that comprehensions will be valued, so do not use them blindly, **always use these tools as an helper!**

Binômes and groups

The whole project is carried out in binômes (pairs).

The binômes are split into **4 groups,** each containing at most 7 teams. Your group matters for two reasons: it is the audience for your presentations (validation and final), and **project topics must be unique within a**

group (see Phase 1).

If a group has an odd number of students, trinômes (three students) are accepted, and monômes (a single student) as well, as long as the group stays within its maximum of 7 teams (binômes, trinômes and monômes combined). The exact composition therefore depends on the number of students in each group.

On Discord: each binôme will chose its **own dedicated channel**. As soon as your binôme is formed, chose a channel (numbered), and write **the full names of both members**.

Phase 1 — Define your project (Session 1)

During the dedicated 2h30 project-definition slot of the first session, your binôme forms, registers on Discord, and defines its project.

By default, define your own project. Choose a subject based on what you actually want to work on and on your previous knowledge, for example a domain you find interesting, a type of data you would like to learn to handle, or a question you would like to answer. Ownership matters: you will live with this project for the whole course, so **pick something that motivates you!**

A good way to get started is to look for a dataset first, then build a task around it. Browsing the open-data platforms listed below and asking yourself "what could I predict, cluster, or extract from this data?" is often the fastest path from a vague interest to a concrete, well-scoped project.

If you cannot converge on your own idea in time, a list of ready-to-use projects is available as a backup on GitHub. Each comes with a candidate dataset and one or two candidate tasks. **This is a safe fallback, not the default!**

Whichever route you take, the **dataset must meet the following criteria**:

- **Adequate Size:** enough samples for meaningful analysis and model training without demanding excessive computational power. A dataset with **1,000–10,000 samples** is usually ideal for beginners. More is ok, less may be problematic.
- **Completeness:** as complete as possible, with minimal missing values.
- **Easy-to-Use Format:** clear labeling of data points, well-defined columns, and readable metadata are essential for easy understanding and processing.
- **Open License:** freely available for academic use.

You will also **define one or two tasks** based on the dataset. These are what you defend in Session 2.

Suggested Platforms for Open-Access Datasets

If you have difficulty finding a dataset or defining tasks, these resources may help:

- **Kaggle:** `https://www.kaggle.com/datasets`
  - A large, beginner-friendly repository of datasets across various domains (healthcare, finance, NLP, etc.).
  - *Example:* Titanic Survival Prediction Dataset (binary classification with tabular data).
- **Hugging Face Datasets:** `https://huggingface.co/datasets`
  - Particularly good for **text and NLP datasets**; also includes datasets for computer vision, reinforcement learning, etc.

    - *Example:* IMDB Reviews (sentiment analysis with labeled text).
- **OpenNeuro:** `https://openneuro.org/`
    - A great source for **neuroimaging datasets**, especially for students interested in AI in neuroscience.
- **PhysioNet:** `https://physionet.org/`
    - Primarily focused on **physiological and clinical data**, great for students interested in healthcare and bioinformatics projects.

Reserve your topic early (before Session 2)

As soon as you have a project idea, post a one-line title of your project in your binôme's Discord channel. A teacher will validate it. **Do this as early as possible, and well before Session 2.**

The main purpose of this early check is to make sure that **no two binômes in the same group work on the same topic.** Within a group, titles are reserved on a first-come basis, so the earlier you post, the safer your topic.

Phase 2 — Validate your project (Session 2)

The entire second session is dedicated to project validation. Each binôme presents its project and the teachers validate it (or ask for a revision). **This presentation is graded.**

Your validation presentation must cover:

1. **Task(s):** the one or two tasks you propose to solve, stated precisely.
2. **Data:** the dataset, its size, format, license, and any obvious limitations.
3. **Metric(s):** how you will measure success for each task.
4. **Existing methods:** what is already known / commonly done for this kind of task (a short state of the art).

Presentation format

- **10 minutes total, questions included.** Keep an eye on the clock. If a presentation uses the full 10 minutes, we will still allow **one** question.
- You **attend all the other presentations in your group** and are **encouraged to ask questions** at the end of each one.
- If your project is **not validated**, you will have to take a **re-take**, with a **penalty on your grade**.

Evaluation -- Project validation

Criterion	Score	Description
Clarity of the proposal	"-"	Task, data or metric is unclear or missing.
	"="	Task(s), data, and metric(s) are clearly and correctly stated.
	"+"	Proposal is precise, well scoped, and the metric is well justified.

Criterion	Score	Description
Awareness of existing methods	"-"	No relevant existing method is identified.
	"="	Existing / standard methods for the task are correctly identified.
	"+"	A concise, well-organized short state of the art is presented.

Phase 3 — Develop the project through the labs (Session 3 → onward)

From Session 3, every lab first applies its techniques to a toy example (to get hands-on with the models). You will then be expected to apply each lab's techniques to your own project data. In a nutshell, here is what you should expect from the different labs:

- **Supervised Learning lab** → train and evaluate a supervised baseline on your data.
- **Unsupervised Learning lab** → clustering / dimensionality reduction / visualization of your data.
- **Deep Learning lab** → a first neural network baseline on your data.
- **Foundation Models lab** → represent your data with a pretrained foundation model (embeddings), and compare with your earlier baselines.
- **Adaptation lab** → adapt or improve a model for your specific task.

Keep track of what you try at each step: this running record will be useful in your final deliverable!

Phase 4 — Final deliverable and presentation

The course ends with sessions dedicated to project work, where you develop, run your final experiments, and consolidate your results. You then present your work in the **final presentation**.

What to present

Present your attempts at solving your task(s). Because the presentation is timed, you will not have time to show everything — and you should not try to. Instead, **start from a clear definition of your task**, then focus on the parts you find most interesting: the techniques that gave you the most insight into your data, achieved the best performance, or are the most established and relevant.

- You may use **any AI tool** introduced in class (e.g., UMAP/t-SNE/PCA for visualization, neural networks, foundation models, etc.).
- The evaluation **emphasizes comprehension**, not the objective performance of your solution! So be pedagogical, and not focused on results.

Presentation format

- **15 minutes total, questions included.**
- **Do not present everything.** Focus on your task definition and on the most interesting things you have to show.
- **Prepare and present a slideshow.**

Good practices

- **Do:** develop a strategy, test hypotheses, and validate them. Extensive experimenting is encouraged!
- **Avoid:** implementations without clear motivation.

Evaluation -- Final deliverable

Criterion	Score	Description
1. Relevance of Overall Strategy	"-"	Experiments lack a clear or consistent approach.
	"="	Experiments follow a logical and consistent approach aligned with the given objectives.
	"+"	Strategy for exploring various AI techniques was particularly effective.
2. Description of Experiments and Results	"-"	The presentation does not clarify which experiments were conducted.
	"="	The presentation is clear and detailed enough to allow for reproduction of the results (clear and comprehensive description of training hyperparameters and architecture).
	"+"	The exploration of AI approaches and hyperparameters is particularly precise and adapted to the chosen dataset.
3. Creativity	"-"	The implementations are limited to the approaches presented in the labs.
	"="	The proposed implementations go beyond the labs, with original techniques, new foundation models, or custom models trained from scratch.
	"+"	The proposed implementations are highly original and creative (involving recent literature research, adoption of new tools and techniques, etc.).